

INTIMATE SUBSPACES, BLOCK FILTERS, AND THE DIAGONALIZATION OF MAXIMAL PROJECTION FILTERS

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ABSTRACT. Fix an orthonormal basis of a separable Hilbert space and call the closed spans of its subsets *blocks*. We prove a zero–cofinite dichotomy answering a question of Fred Galvin: a closed subspace of infinite codimension misses some infinite block entirely, so every closed subspace, restricted to a suitable infinite block, is either everything but finitely much or nothing at all. Using it we show that the filter of closed subspaces generated by the blocks of an ultrafilter $\mathcal{U}$ on $\mathbb{N}$, together with the finite-codimension subspaces, is a *maximal* filter in the projection lattice (a *pluf*, in the terminology of the companion paper) if and only if $\mathcal{U}$ is selective. For non-selective $\mathcal{U}$ the block filter is properly extendable; we prove that its extensions cannot be steered away from the witness subspace by any blocker built from disjointly supported vectors, nor by any subspace meeting a single piece-block in dimension two or more, and we conjecture that the witness subspace belongs to *every* maximalization — a rigidity phenomenon with no Boolean analogue. We then develop the theory of *intimate* subspaces: those meeting, for every two-coloring of the basis, at least one of the two color blocks nontrivially. A construction of Gowers, reproduced here with proof, shows intimacy strictly exceeds its obvious sufficient conditions; our dichotomy shows that example has, in a precise sense, the only possible shape. The pay-off is a characterization: a nonprincipal pluf is diagonalizable over a basis if and only if every member is intimate for that basis. Consequently non-diagonalizability is witnessed by a single member, the transfinite machinery of the companion paper collapses to a one-witness reduction, and the question of ZFC examples localizes to a concrete question about the continuous masa.

1. INTRODUCTION

Throughout, $\mathcal{H}$ is a separable infinite-dimensional Hilbert space with a fixed orthonormal basis $(e_n)_{n \in \mathbb{N}}$; for $S \subseteq \mathbb{N}$ we write $\mathcal{H}_S = \overline{\text{span}}\{e_n : n \in S\}$ and call such subspaces *blocks*. A *pluf* is a maximal filter in the lattice $\mathcal{P}(\mathcal{H})$ of closed subspaces; the companion paper [5] develops the general theory. Here we study the interaction of plufs with a basis, organized around three results.

First, a dichotomy (§2). Gowers attributed to Fred Galvin the question: must a closed subspace meeting every infinite block have finite codimension? Theorem 2.1 answers it in the sharp form that a subspace of infinite codimension misses some infinite block *entirely*, completing a sketch of Gowers. Relativized (Corollary 2.3), this says the restriction of any closed subspace to a suitable infinite sub-block is either cofinite-dimensional or zero — no middle ground survives passage to a sub-block.

Second, a characterization of maximality (§3). An ultrafilter $\mathcal{U}$ on $\mathbb{N}$ generates, together with the finite-codimension subspaces, a filter $\Phi(\mathcal{U})$ in $\mathcal{P}(\mathcal{H})$. Theorem 3.3: $\Phi(\mathcal{U})$ is a pluf if and only if $\mathcal{U}$ is selective. The proof combines Corollary 2.3 with Mathias’s theorem on happy families; necessity is witnessed by an explicit subspace W built from a partition without selectors. We then study the pluf extensions of $\Phi(\mathcal{U})$ in the non-selective case and prove two rigidity results (Proposition 3.5, Lemma 3.4): the natural candidates for a member separating an extension from W all fail, for reasons that appear structural rather than accidental. We

conjecture that W belongs to every pluf extension of $\Phi(\mathcal{U})$ (Conjecture 3.6) — maximalization in the projection lattice, unlike in a Boolean algebra, would then be canalized through specific transverse subspaces.

Third, intimacy and diagonalization (§§4–5). Call W *intimate* (for the fixed basis) if for every $A \subseteq \mathbb{N}$, $W \cap \mathcal{H}_A \neq 0$ or $W \cap \mathcal{H}_{\mathbb{N} \setminus A} \neq 0$. Subspaces of finite codimension in an infinite block, and their enlargements, are intimate; Gowers’s construction (Theorem 4.2) shows that these do not exhaust intimacy. Corollary 2.3 then shows the example is not an accident of the construction: *any* intimate subspace beyond the trivial examples must have dense zero-set (every infinite set contains an infinite S with $W \cap \mathcal{H}_S = 0$) while its zero-set contains no complementary pair (Proposition 4.5) — exactly the shape Gowers built.

No characterization of intimacy is needed for the application: what plufs consume is its *failure*, one witness at a time.

Theorem 1.1 (= Theorem 5.2). *A nonprincipal pluf π is diagonalizable over a basis $\mathcal{B}$ — meaning $\{S : \mathcal{H}_S^\mathcal{B} \in \pi\}$ is an ultrafilter — if and only if every member of π is $\mathcal{B}$ -intimate.*

Since non-intimacy of a single member survives every enlargement of a filter, the diagonalizability of all extensions of a family is decided by the family itself (Proposition 5.5). Theorem 5.6 builds such a family under CH: a non-diagonalizable nonprincipal pluf obtained without spectral theory, and without carrying a maximalization through the recursion. Section 6 assembles what is known toward removing CH, including the observation that countable nested families can never witness the reduction, and the identification of the continuous masa as the natural candidate.

2. THE ZERO-COFINITE DICHOTOMY

Theorem 2.1 (Galvin’s question; completing a sketch of Gowers). *Let W be a closed subspace of $\mathcal{H}$ of infinite codimension. Then there is an infinite $S \subseteq \mathbb{N}$ with $W \cap \mathcal{H}_S = 0$.*

Proof. Let $U = W^\perp$, so $\dim U = \infty$. For each N , the subspace $\{u \in U : u_n = 0 \text{ for } n \leq N\} = \ker(P_{[1,N]}|_U)$ has codimension at most N in U , hence is nonzero. Recursively choose $N_0 = 0 < n_1 < N_1 < n_2 < N_2 < \dots$ and unit vectors $u_j \in U$ as follows: u_j vanishes on $[1, N_{j-1}]$; $n_j > N_{j-1}$ is a coordinate with $\delta_j := |u_j(n_j)| > 0$; and $N_j > n_j$ is large enough that $(\sum_{n > N_j} |u_j(n)|^2)^{1/2} < \delta_j 2^{-j-2}$.

Set $S = \{n_1 < n_2 < \dots\}$ and let $x \in W \cap \mathcal{H}_S$. Fix j . For $i < j$ we have $n_i \leq N_i \leq N_{j-1}$, so $u_j(n_i) = 0$; hence

$$0 = \langle x, u_j \rangle = x_{n_j} \overline{u_j(n_j)} + \sum_{i > j} x_{n_i} \overline{u_j(n_i)}.$$

For $i > j$, $n_i > N_{i-1} \geq N_j$, so Cauchy–Schwarz bounds the sum by $\|x\| \delta_j 2^{-j-2}$, giving $|x_{n_j}| \leq 2^{-j-2} \|x\|$. Then $\|x\|^2 = \sum_j |x_{n_j}|^2 \leq \|x\|^2 \sum_{j \geq 1} 4^{-j-2} = \frac{1}{48} \|x\|^2$, so $x = 0$. $\square$

Remark 2.2. Gowers’s sketch (row-echelon form for a spanning family of $W^\perp$, distinguished columns with rapidly decreasing tails) is exactly the recursion above; the convergence book-keeping is the only addition. We have not found the statement in the literature; it has the flavor of a Bessaga–Pełczyński selection and may be known to Banach-space specialists.

Corollary 2.3 (Relativized dichotomy). *For every closed W and every infinite $S_0 \subseteq \mathbb{N}$: either $W \cap \mathcal{H}_{S_0}$ has finite codimension in $\mathcal{H}_{S_0}$, or there is an infinite $S \subseteq S_0$ with $W \cap \mathcal{H}_S = 0$.*

Proof. Apply Theorem 2.1 inside $\mathcal{H}_{S_0}$ to the subspace $W \cap \mathcal{H}_{S_0}$. $\square$

Thus the conceivable middle behaviors — infinite dimension and infinite codimension at once — can always be destroyed by passing to a sub-block. Write

$$\mathcal{Z}_W = \{S \in [\mathbb{N}]^\infty : W \cap \mathcal{H}_S = 0\}, \quad \mathcal{C}_W = \{S \in [\mathbb{N}]^\infty : \text{codim}_{\mathcal{H}_S}(W \cap \mathcal{H}_S) < \infty\};$$

both are closed under infinite subsets, and Corollary 2.3 says $\mathcal{Z}_W \cup \mathcal{C}_W$ is dense in $([\mathbb{N}]^\infty, \subseteq)$.

3. BLOCK FILTERS AND SELECTIVITY

Definition 3.1. For an ultrafilter $\mathcal{U}$ on $\mathbb{N}$, let $\Phi(\mathcal{U})$ be the filter in $\mathcal{P}(\mathcal{H})$ generated by $\{\mathcal{H}_S : S \in \mathcal{U}\}$ together with all closed subspaces of finite codimension.

$\Phi(\mathcal{U})$ is proper: a generating intersection has the form $\mathcal{H}_S \cap V$ with $S \in \mathcal{U}$ and $\text{codim } V < \infty$, which is infinite-dimensional. Two criteria are immediate:

Lemma 3.2. (a) $W \in \Phi(\mathcal{U})$ iff $\text{codim}_{\mathcal{H}_S}(W \cap \mathcal{H}_S) < \infty$ for some $S \in \mathcal{U}$. (b) W generates a proper filter with $\Phi(\mathcal{U})$ iff $\dim(W \cap \mathcal{H}_S) = \infty$ for every $S \in \mathcal{U}$.

Proof. (a) If $\text{codim}_{\mathcal{H}_S}(W \cap \mathcal{H}_S) = d < \infty$, then $V = (W \cap \mathcal{H}_S) \oplus \mathcal{H}_{\mathbb{N} \setminus S}$ has codimension d and $\mathcal{H}_S \cap V \subseteq W$. Conversely if $\mathcal{H}_S \cap V \subseteq W$ with $\text{codim } V < \infty$ then $\text{codim}_{\mathcal{H}_S}(W \cap \mathcal{H}_S) \leq \text{codim } V$. (b) X meets every finite-codimension subspace nontrivially iff $\dim X = \infty$; apply this to $X = W \cap \mathcal{H}_S$ within the generating intersections. $\square$

Recall that $\mathcal{U}$ is *selective* (Ramsey) iff every partition of $\mathbb{N}$ into sets not in $\mathcal{U}$ admits a selector in $\mathcal{U}$, iff $\mathcal{U}$ is minimal in the Rudin–Keisler order, and that Mathias’s theorem gives: for selective $\mathcal{U}$ and analytic $\mathcal{A} \subseteq [\mathbb{N}]^\infty$ there is $S \in \mathcal{U}$ with $[S]^\infty \subseteq \mathcal{A}$ or $[S]^\infty \cap \mathcal{A} = \emptyset$ [9].

Theorem 3.3. $\Phi(\mathcal{U})$ is a pluf if and only if $\mathcal{U}$ is selective.

Proof. Sufficiency. Let $\mathcal{U}$ be selective and W closed. The set $\{S : W \cap \mathcal{H}_S \neq 0\}$ is analytic (it is the projection of the closed set $\{(S, x) : x \in W, \|x\| = 1, \text{supp } x \subseteq S\}$). By Mathias, some $S \in \mathcal{U}$ has $[S]^\infty$ inside or disjoint from it. In the disjoint case $W \cap \mathcal{H}_S = 0$: the block $\mathcal{H}_S \in \Phi(\mathcal{U})$ blocks W . In the inside case no infinite subset of S lies in $\mathcal{Z}_W$, so Corollary 2.3 forces $S \in \mathcal{C}_W$, and $W \in \Phi(\mathcal{U})$ by Lemma 3.2(a). Every W being decided, $\Phi(\mathcal{U})$ is maximal (by the maximality criterion of [5], Lemma 2.1: a filter is maximal iff every subspace is a member or meets some member trivially).

Necessity. Let $\mathcal{U}$ be non-selective and fix a partition $\mathbb{N} = \bigsqcup_k A_k$ with no $A_k \in \mathcal{U}$ and no selector in $\mathcal{U}$. Put $f_k = \sum_{n \in A_k} 2^{-n} e_n$ and $W = \{f_k : k\}^\perp$. The restrictions $P_S f_k$, over k with $A_k \cap S \neq \emptyset$, have pairwise disjoint supports, hence are linearly independent; so for every S ,

$$\dim(W \cap \mathcal{H}_S) = \sum_k (|A_k \cap S| - 1)^+, \quad \text{codim}_{\mathcal{H}_S}(W \cap \mathcal{H}_S) = \#\{k : A_k \cap S \neq \emptyset\}$$

(with the convention $\infty - 1 = \infty$). For $S \in \mathcal{U}$: if the codimension were finite, S would lie in a finite union of the A_k up to a finite set, putting some A_k in $\mathcal{U}$; so $W \notin \Phi(\mathcal{U})$ by Lemma 3.2(a). If the dimension were finite, deleting finitely many points of S would leave a set meeting each A_k at most once — a selector in $\mathcal{U}$; so $\dim(W \cap \mathcal{H}_S) = \infty$ for all $S \in \mathcal{U}$, and by Lemma 3.2(b), W is addable. Addable and unadded: $\Phi(\mathcal{U})$ is not maximal. $\square$

We now examine the extensions of $\Phi(\mathcal{U})$ in the non-selective case. By the maximality criterion, a pluf extension of $\Phi(\mathcal{U})$ omits W if and only if it has a member R with $R \cap W = 0$; and every member of every extension meets every member of $\Phi(\mathcal{U})$, i.e. satisfies $\dim(R \cap \mathcal{H}_S) = \infty$ for all $S \in \mathcal{U}$ (Lemma 3.2(b)). Call such an R an *addable blocker* of W . One might expect addable blockers to abound, as they do in the Boolean setting, where a filter omitting a set always extends to an ultrafilter omitting it (adjoin the complement). The next two results show that in $\mathcal{P}(\mathcal{H})$ the natural candidates all fail, in a way that suggests the failure is structural.

Lemma 3.4 (Blockers are thin on every piece). *If R is closed with $R \cap W = 0$, then $\dim(R \cap \ell^2(A_k)) \leq 1$ for every k .*

Proof. If $\dim(R \cap \ell^2(A_k)) \geq 2$, the single linear condition $\langle \cdot, f_k \rangle = 0$ cuts it in codimension at most one, leaving a nonzero $v \in R \cap \ell^2(A_k) \cap f_k^\perp$; and $v \perp f_j$ for $j \neq k$ automatically (disjoint supports), so $0 \neq v \in R \cap W$. $\square$

Proposition 3.5 (No disjointly supported blocker). *Let R be the closed span of a family (ρ_i) of nonzero vectors with pairwise disjoint supports $\sigma_i = \text{supp } \rho_i$, and suppose $R \cap W = 0$. Then there is $S \in \mathcal{U}$ with $R \cap \mathcal{H}_S = 0$. In particular R is not addable, and no pluf extension of $\Phi(\mathcal{U})$ omitting W has a member of this form.*

Proof. Since the supports are disjoint, the normalized ρ_i are orthonormal, so every $x \in R$ has a unique expansion $x = \sum_i c_i \rho_i$, and the restriction of x to σ_i is $c_i \rho_i|_{\sigma_i}$. Hence $x \in \mathcal{H}_S$ iff for every i either $c_i = 0$ or $\sigma_i \subseteq S$:

$$R \cap \mathcal{H}_S = \overline{\text{span}}\{\rho_i : \sigma_i \subseteq S\}.$$

It therefore suffices to produce $S \in \mathcal{U}$ containing no σ_i , i.e. a set $D \notin \mathcal{U}$ meeting every σ_i .

Let D_s be the set of points carrying singleton supports. If two of them, $n \neq n'$, lay in one piece A_k , then $e_n, e_{n'} \in R$ (scalar multiples of the two generators) and the vector $2^{-n'}e_n - 2^{-n}e_{n'}$ would lie in $R \cap W$: it is orthogonal to f_k by the choice of coefficients and to every other f_j by disjointness of supports. So $R \cap W = 0$ forces $|D_s \cap A_k| \leq 1$ for all k : D_s is a selector of the partition, hence $D_s \notin \mathcal{U}$.

For each i with $|\sigma_i| \geq 2$, choose two distinct points $p_i, q_i \in \sigma_i$. The sets $P = \{p_i\}$ and $Q = \{q_i\}$ are disjoint, so at most one of them lies in the ultrafilter; let $D_1 \in \{P, Q\}$ with $D_1 \notin \mathcal{U}$. Then $D = D_s \cup D_1$ is a union of two non-members, hence $D \notin \mathcal{U}$, and $S = \mathbb{N} \setminus D \in \mathcal{U}$. Every σ_i meets D (singletons through D_s , the rest through D_1), so no $\sigma_i \subseteq S$ and $R \cap \mathcal{H}_S = 0$. $\square$

Lemma 3.4 says a blocker can carry at most a line's worth of each piece-block; Proposition 3.5 says that assembling such lines (or any disjointly supported family, aligned with the pieces or not) never yields addability, because the ultrafilter always contains the complement of a transversal of the supports. Blockers, if they exist at all, must therefore couple infinitely many pieces through vectors with overlapping supports, in such a way that $R \cap \mathcal{H}_S$ is sustained by cancellation rather than by whole generators. Three further constraints on such an R are proved in [7], §6: its intersection with the block of any m pieces has dimension at most m ; its total support lies in $\mathcal{U}$; and inside every $\mathcal{H}_S$, $S \in \mathcal{U}$, it contains vectors meeting infinitely many pieces.

Conjecture 3.6 (Rigidity). *For non-selective $\mathcal{U}$ and the witness subspace W of Theorem 3.3, no addable blocker exists: every closed R with $R \cap W = 0$ satisfies $\dim(R \cap \mathcal{H}_S) < \infty$ for some $S \in \mathcal{U}$. Equivalently, W belongs to every pluf extension of $\Phi(\mathcal{U})$.*

Remark 3.7. The conjecture, if true, exhibits a *forced member*: a subspace outside a filter that nevertheless belongs to every maximalization of it. Nothing of the kind occurs in a Boolean algebra, where the extension space of a non-maximal filter always splits over any undecided set. In $\mathcal{P}(\mathcal{H})$ maximalization would then not be the free completion it is classically, but canalized through transverse subspaces determined by the combinatorics of $\mathcal{U}$. Whether unique maximalization characterizes selectivity the conjecture leaves open, since extensions containing W may still differ elsewhere. An earlier version of this paper asserted incompatible maximalizations for every non-selective $\mathcal{U}$; that assertion is withdrawn.

Remark 3.8 (Formal verification). Sections 2–6 have been formalized and machine-checked in Lean 4 with Mathlib (toolchain v4.28.0): Theorem 2.1 and Corollary 2.3 (the gliding-hump recursion and its relativization, the latter proved by re-running the recursion inside a block rather than by transport along an enumeration); the density clause of Proposition 4.5; Lemma 3.2, both criteria, and properness of $\Phi(\mathcal{U})$; Theorem 3.3 in both directions — necessity in full, sufficiency parametrized by the conclusion of Mathias’s theorem for the subspace in question, which is the quarantined classical import; Lemma 3.4, Proposition 3.5, and the construction of W and the blocks; Theorem 4.2 in all three clauses (intimacy, the finite-block dimension bound, and the absence of finite relative codimension); and Theorem 5.2 together with Corollary 5.4(a), stated against a named package of the lattice facts consumed from [5] — upward closure, closure under meets, properness, membership of the finite-codimension subspaces, and the maximality criterion — so that the mechanized statements are theorems of ZFC. The artifacts compile with no unproved obligations, and every statement’s axiom footprint is exactly `propext`, `Classical.choice`, `Quot.sound`. Three by-products of the mechanization are worth recording: in Proposition 3.5 neither the nonemptiness of the pieces nor their covering $\mathbb{N}$ is used; the finite-block bound of Theorem 4.2 admits the sharper proof of Remark 4.3 below; and the extension formulation of Corollary 5.4(a) requires its quantifier restricted to closed subspaces, since the filter generated by the blocks of $\mathcal{U}$ and the finite-codimension subspaces contains subspaces that are not closed. Theorem 5.6 is verified with the continuum hypothesis taken as an equinumerosity hypothesis, by a route this paper itself makes available: rather than running the three-case recursion of [5], the mechanization builds the directed family and appeals to Proposition 5.5, so that maximality, nonprincipality and non-diagonalizability all follow from results already proved; the blocking subspace at each stage is then a line, and the three cases collapse to one. Proposition 5.5 is verified as printed, quantified over all orthonormal bases indexed by $\mathbb{N}$ — which required the basis-relative form of Theorem 5.2, itself established at an arbitrary Hilbert basis of an arbitrary real Hilbert space — and Proposition 6.1 likewise, its coordinatization resting on a countable orthogonal decomposition adapted to the chain. The verification artifacts will be archived with the final version of this paper.

Remark 3.9. Since selective ultrafilters need not exist (they are absent from the random real model of Kunen), it is consistent that *no* basis-generated pluf exists, although plufs always exist by Zorn’s lemma. This bears on the strength of the existence of nonprincipal plufs: the basis route cannot establish it from the Boolean Prime Ideal theorem, since selectivity exceeds BPI. Whether BPI implies the existence of nonprincipal plufs we leave open; the natural compactness reduction fails because maximality in $\mathcal{P}(\mathcal{H})$ (unlike in a Boolean algebra) is an infinitary disjunction over the blockers of each subspace.

4. INTIMATE SUBSPACES

Definition 4.1. A closed subspace $W \subseteq \mathcal{H}$ is *intimate* (for the basis (e_n)) if for every $A \subseteq \mathbb{N}$, at least one of $W \cap \mathcal{H}_A$, $W \cap \mathcal{H}_{\mathbb{N} \setminus A}$ is nonzero.

Equivalently: W is non-intimate iff for some two-coloring of the basis, the coordinate projections of both colors are injective on W — iff W is, in the incidence sense, a *bi-graph* over some splitting. Intimacy is thus a partition property of W against the basis. Trivial examples: any W containing a subspace of finite codimension in an infinite block $\mathcal{H}_B$ is intimate, since for any coloring one color meets B infinitely and finite codimension survives to that side. The trivial examples do not exhaust intimacy. Gowers’s construction follows, with one typographical repair (the display of the 2-to-1 map) and a concrete choice of the auxiliary vectors.

Theorem 4.2 (Gowers, personal communication, c. 2000). *There is an intimate closed subspace X such that $\dim(X \cap \mathcal{H}_C) \leq |C|/2$ for every finite $C \subseteq \mathbb{N}$. Consequently X contains no subspace of finite codimension in any infinite block; the trivial examples do not exhaust intimacy.*

Proof. Let $w = (1, \frac{1}{2}, \frac{1}{3}, \dots) \in \ell^2$ and choose vectors $u_n \in \ell^2$ with first nonzero coordinate at n and $\langle u_n, w \rangle = 0$; concretely,

$$u_n = e_n - \frac{n+1}{n} e_{n+1} \quad \left(\langle u_n, w \rangle = \frac{1}{n} - \frac{n+1}{n} \cdot \frac{1}{n+1} = 0 \right).$$

Duplicate: $v_{n,2i-1} = v_{n,2i} = u_{n,i}$, and set $X = \{v_n : n\}^\perp$. Note $e_{2i-1} - e_{2i} \in X$ for every i .

Intimacy. Let $\mathbb{N} = A \sqcup B$. If some pair $\{2i-1, 2i\}$ lies in A , then $e_{2i-1} - e_{2i} \in X \cap \mathcal{H}_A$; similarly for B . Otherwise every pair splits, so $A = \{a_1 < a_2 < \dots\}$ with $a_i \in \{2i-1, 2i\}$; the vector $x = \sum_i w_i e_{a_i}$ lies in ℓ^2 and satisfies $\langle x, v_n \rangle = \sum_i w_i v_{n,a_i} = \sum_i w_i u_{n,i} = \langle w, u_n \rangle = 0$, so $0 \neq x \in X \cap \mathcal{H}_A$ (and symmetrically $X \cap \mathcal{H}_B \neq 0$: in the split case X meets *both* sides).

The dimension bound. Let $|C| = n$ and set $I = \{\lceil j/2 \rceil : j \in C\}$, so $|I| \geq n/2$ (the map $j \mapsto \lceil j/2 \rceil$ is at worst 2-to-1). For $i \in I$, the restriction $v_i|_C$ is nonzero: a witness $j \in C \cap \{2i-1, 2i\}$ has $v_{i,j} = u_{i,i} \neq 0$. Order $I = \{i_1 < i_2 < \dots\}$; since v_i vanishes at positions $< 2i-1$, the row $v_{i_{k'}}|_C$ vanishes at the witness position $j_k \leq 2i_k$ of every earlier row ($2i_k < 2i_{k'} - 1$): the rows are in echelon form, hence linearly independent, imposing at least $n/2$ independent conditions on $\mathcal{H}_C$. So $\dim(X \cap \mathcal{H}_C) \leq n/2$.

Finally, finite codimension d inside an infinite block $\mathcal{H}_S$ would force $\dim(X \cap \mathcal{H}_C) \geq |C| - d$ for finite $C \subseteq S$, contradicting the bound for $|C| > 2d$. $\square$

Remark 4.3 (The finitely supported part of X). The dimension bound admits a sharper argument, which identifies the intersection exactly. Membership in X is the recursion $s_n = \frac{n+2}{n+1} s_{n+1}$ on the pair sums $s_n = x_{2n} + x_{2n+1}$ (0-indexed), whose coefficients never vanish; so if $x \in X$ has finite support, running the recursion downward from a vanishing tail forces *every* pair sum to vanish. Thus $X \cap \mathcal{H}_C = \{x : x_{2k} = -x_{2k+1}\}$ on complete pairs and $x = 0$ elsewhere: the finitely supported part of X is exactly the span of the pair differences $e_{2k} - e_{2k+1}$, and $\dim(X \cap \mathcal{H}_C)$ equals the number of complete pairs of C , whence $2 \dim(X \cap \mathcal{H}_C) \leq |C|$, with equality iff C is a union of pairs.

Remark 4.4 (The example is finely tuned to its basis). With the concrete u_n above, $\{u_n\}^\perp = \mathbb{R}w$ exactly (solve the recursion $x_n = \frac{n+1}{n} x_{n+1}$). Hence for the evens/odds splitting — where every pair splits — the intersections $X \cap \mathcal{H}_{\text{odd}}$ and $X \cap \mathcal{H}_{\text{even}}$ are each *one-dimensional*, spanned by the twisted copies of w . So X is intimate only in the nonvanishing sense of Definition 4.1; its “infinite-dimensional intimacy” fails, and X is far from addable to block filters. For the applications below, nonvanishing intimacy is exactly the right notion.

Corollary 2.3 explains why any counterexample had to look like this.

Proposition 4.5 (The shape of nontrivial intimacy). *W is intimate iff $\mathcal{Z}_W$ contains no complementary pair $\{A, \mathbb{N} \setminus A\}$. If in addition W has infinite codimension in every block it nearly fills — i.e. $\mathcal{C}_W = \emptyset$ — then $\mathcal{Z}_W$ is dense: every infinite set contains an infinite S with $W \cap \mathcal{H}_S = 0$. Gowers’s X satisfies both conditions.*

Proof. The first sentence is the definition. The second is Corollary 2.3. For X : $\mathcal{C}_X = \emptyset$ by the dimension bound, as in the last step of Theorem 4.2. $\square$

Thus the “hard case” of intimacy is precisely: a dense zero-set avoiding complementary pairs. This is a purely combinatorial specification, and it also explains why such W never obstruct Theorem 3.3: density of $\mathcal{Z}_W$ is exactly what Mathias’s theorem consumes. Whether

the notion admits a structure theory in the style of the Gowers dichotomies — via the exact Ramsey principle of Rosendal [11] and the Gowers-space framework of de Rancourt [4], in which the relevant pigeonhole principle visibly fails and the truth should be game-theoretic — we leave as a program; the next section shows the pluf application does not wait on it.

5. DIAGONALIZABLE PLUFS

Definition 5.1. A pluf π is *diagonalizable* via an orthonormal basis $\mathcal{B}$ if $\mathcal{U}_\pi^\mathcal{B} := \{S : \mathcal{H}_S^\mathcal{B} \in \pi\}$ is an ultrafilter on the index set of $\mathcal{B}$. (π is *basic* if $\pi = \Phi(\mathcal{U})$ for some basis and ultrafilter.)

By [5], Lemma 2.1, a filter π is maximal iff every subspace is a member or meets some member trivially; and every nonprincipal pluf contains every finite-codimension subspace. We use both freely.

Theorem 5.2. *A nonprincipal pluf π is diagonalizable via $\mathcal{B}$ if and only if every member of π is $\mathcal{B}$ -intimate.*

Proof. ($\Rightarrow$) If $R \in \pi$ satisfies $R \cap \mathcal{H}_A = R \cap \mathcal{H}_{A^c} = 0$ then neither $\mathcal{H}_A$ nor $\mathcal{H}_{A^c}$ can join π , so the trace is not an ultrafilter.

($\Leftarrow$) Given A , suppose $\mathcal{H}_A, \mathcal{H}_{A^c} \notin \pi$. By maximality pick $R_1, R_2 \in \pi$ with $R_1 \cap \mathcal{H}_A = 0$ and $R_2 \cap \mathcal{H}_{A^c} = 0$; then $R_1 \cap R_2 \in \pi$ is not $\mathcal{B}$ -intimate. So the trace decides every A ; it never contains both A and A^c (their blocks meet trivially); and it handles finite A correctly, since $\mathcal{H}_{A^c}$ then has finite codimension and belongs to π . Filter properties of the trace follow from those of π via $\mathcal{H}_S \cap \mathcal{H}_{S'} = \mathcal{H}_{S \cap S'}$. $\square$

Remark 5.3. The proof of Theorem 5.2 uses only the lattice identities $\mathcal{H}_S \cap \mathcal{H}_{S'} = \mathcal{H}_{S \cap S'}$, monotonicity of $S \mapsto \mathcal{H}_S$, and the triviality of $\mathcal{H}_A \cap \mathcal{H}_{A^c}$, together with membership of the finite-codimension subspaces. It therefore holds verbatim for an arbitrary orthonormal basis of an arbitrary real Hilbert space, with the index set of the basis in place of $\mathbb{N}$; separability plays no role.

Corollary 5.4. (a) π is diagonalizable iff $\pi \supseteq \Phi(\mathcal{U})$ for some basis and ultrafilter; in particular diagonalizable nonprincipal plufts exist in ZFC. (b) π is basic iff it is diagonalizable via a selective ultrafilter (Theorem 3.3). (c) Every diagonalizable pluf π has S_π a singleton — the state-space face of [5], §3 — by Marcus–Spielman–Srivastava; hence the round-slice property for every ellipsoid. Consequently the ample pluf of [5], Theorem 5.6 is non-diagonalizable, and the lattice analogue of Anderson’s conjecture [2] fails under CH.

Non-intimacy of a single member survives every enlargement of a filter, so non-diagonalizability is decided by any family the pluf contains.

Proposition 5.5 (One-witness reduction). *Let $\mathcal{R}$ be a downward directed family of closed subspaces, all finite intersections infinite-dimensional, with $\bigcap \mathcal{R} = 0$, such that every orthonormal basis $\mathcal{B}$ admits some $R \in \mathcal{R}$ that is not $\mathcal{B}$ -intimate. Then every pluf extending $\mathcal{R}$ (such exist, by Zorn) is nonprincipal and non-diagonalizable.*

Proof. Nonprincipality: $\bigcap \mathcal{R} = 0$ rules out $\bigcap \pi \neq 0$ by [5], Corollary 2.3. Non-diagonalizability via any $\mathcal{B}$: the witness $R \in \mathcal{R} \subseteq \pi$ and Theorem 5.2. $\square$

Under CH such a family can be built one basis at a time. The construction needs no spectral theory, and no maximalization is carried through the recursion: only the family is built, and Proposition 5.5 supplies the pluf.

Theorem 5.6. *Assume CH. There is a nonprincipal pluf π that is diagonalizable via no orthonormal basis.*

Proof. Enumerate the orthonormal bases as $\{\mathcal{B}_i : i < \omega_1\}$ and the nonzero vectors of $\mathcal{H}$ as $\{v_i : i < \omega_1\}$; there are $2^{\aleph_0} = \aleph_1$ of each. We construct closed subspaces R_i , $i < \omega_1$, so that, writing G_i for the family of finite intersections of $\{R_j : j < i\}$ (with $G_0 = \{\mathcal{H}\}$):

- (i) $\dim(R_i \cap g) = \infty$ for every $g \in G_i$;
- (ii) R_i is not $\mathcal{B}_i$ -intimate;
- (iii) $R_i \cap \mathbb{R}v_i = 0$.

Suppose R_j is given for $j < i$. By (i) at earlier stages, G_i is a countable downward directed family of infinite-dimensional subspaces, so it has a decreasing cofinal chain $h_1 \supseteq h_2 \supseteq \dots$. Write $N = \mathbb{R}v_i$ and $\mathcal{B}_i = (e_\beta^{(i)})$. Choose unit vectors $w_n \in h_n$ and thresholds $N_1 < N_2 < \dots$ recursively so that:

- (a) $w_n \perp e_\beta^{(i)}$ for all $\beta \leq N_n$;
- (b) $w_n \perp w_j$ and $w_n \perp P_{N^\perp} w_j$ for $j < n$, and $w_n \notin N$;
- (c) w_n has at least two nonzero $\mathcal{B}_i$ -coordinates; fix two, $a_n < b_n$, and take $N_{n+1} > b_n$.

Every condition but the two nonvanishing ones is a finite set of linear constraints, carving a finite-codimension subspace C_n out of the infinite-dimensional h_n ; so C_n is infinite-dimensional. Since N is a line, $C_n \not\subseteq N$, which gives $w_n \notin N$. And C_n contains a vector with two nonzero coordinates, since a subspace all of whose vectors are multiples of single basis vectors has dimension at most one: given independent $u, v \in C_n$, one of $u, v, u + v$ will do.

Set $A = \{a_n : n\}$ and $R_i = \overline{\text{span}}\{w_n\}$. The w_n are orthonormal by (b), so each $x \in R_i$ has an expansion $x = \sum c_n w_n$. If L is least with $c_L \neq 0$ then, since w_m vanishes on coordinates $\leq N_m$ and $a_L, b_L \leq N_{L+1} \leq N_m$ for $m > L$,

$$x_{a_L} = c_L w_L(a_L) \neq 0, \quad x_{b_L} = c_L w_L(b_L) \neq 0.$$

As $a_L \in A$ and $b_L \notin A$, no nonzero $x \in R_i$ lies in $\mathcal{H}_A$ or in $\mathcal{H}_{A^c}$: $R_i \cap \mathcal{H}_A = R_i \cap \mathcal{H}_{A^c} = 0$, which is (ii). The vectors $y_n = P_{N^\perp} w_n$ are nonzero by (b), and pairwise orthogonal, since for $j < n$

$$\langle y_n, y_j \rangle = \langle w_n, P_{N^\perp} y_j \rangle = \langle w_n, y_j \rangle = 0;$$

so $P_{N^\perp} x = \sum c_n y_n$ vanishes only for $x = 0$, giving $R_i \cap N = 0$, which is (iii). Finally $w_n \in h_n \subseteq h_j$ for $n \geq j$, so $\dim(R_i \cap h_j) = \infty$ for every j , and cofinality of the chain in G_i gives (i) at every later stage.

Let $\mathcal{R}$ be the family of finite intersections of the R_i . It is downward directed, and all its members are infinite-dimensional by (i). Every nonzero vector of $\mathcal{H}$ is some v_i , and $R_i \cap \mathbb{R}v_i = 0$, so $\bigcap \mathcal{R} = 0$. Every orthonormal basis is some $\mathcal{B}_i$, and $R_i \in \mathcal{R}$ is not $\mathcal{B}_i$ -intimate. Proposition 5.5 applies. $\square$

The ZFC question of [5] thus localizes: *exhibit, in ZFC, a single directed family defeating all bases.*

6. TOWARD ZFC

We record the constraints known to us on any witness family for Proposition 5.5.

Proposition 6.1 (Chains never suffice). *Any countable nested family $M_1 \supseteq M_2 \supseteq \dots$ of closed subspaces is simultaneously coordinatizable: there is a basis $\mathcal{B}$ for which every M_k is a block. Blocks are intimate for their own basis; hence every member of the family, and every enlargement of a member, is $\mathcal{B}$ -intimate, and no countable chain (nor any family of countable cofinality, applied to a cofinal chain) witnesses Proposition 5.5.*

Proof. Choose orthonormal bases of $\bigcap_k M_k$, of each difference $M_k \ominus M_{k+1}$, and of $\mathcal{H} \ominus M_1$; their union is a basis making each M_k a block. A block $\mathcal{H}_B$ is intimate: for any splitting $A \sqcup A^c$, one side meets B , and $\mathcal{H}_B \cap \mathcal{H}_A = \mathcal{H}_{B \cap A}$. $\square$

The natural candidate is therefore *wide*: the full continuous chain $\mathcal{R} = \{L^2[0, t] : t \in (0, 1]\}$ in $\mathcal{H} = L^2[0, 1]$, with $\bigcap \mathcal{R} = 0$. It is not simultaneously coordinatizable: the projections $P_{[0, t]}$ generate the continuous masa $L^\infty[0, 1]$, which, being maximal abelian, cannot embed in the atomic masa of any basis. But non-coordinate does not imply non-intimate, and the question stands:

Question 6.2. Is there, for every orthonormal basis $\mathcal{B}$ of $L^2[0, 1]$, some t with $L^2[0, t]$ not $\mathcal{B}$ -intimate? An affirmative answer gives, via Proposition 5.5, a ZFC nonprincipal non-diagonalizable pluf, settling the lattice Anderson conjecture outright.

Three orienting remarks. First, the state-theoretic route does not shortcut the question: the non-diagonalizable pure states of Akemann–Weaver [1] (under CH) and Koszmider [8] (in ZFC) are singular, hence non-regular, and their 1-sets are quantum filters that are *not* closed under meets — they yield no lattice filter (see [3], Lemma 4.3 and the surrounding discussion, and [6] for the positive counterpart: countably additive pure states *do* yield plufs). Second, at measurable dimension the question changes character: no family of coordinate subspaces can defeat all bases there, by the exact dichotomy of [6], and a singular countably additive state annihilates every separable-range projection, so the Lebesgue chain becomes invisible; the correct large-cardinal analogue of Question 6.2 concerns $L^2(\{0, 1\}^\kappa)$. Third, Gowers’s example cuts the other way (Remark 4.4): its intimacy is genuinely basis-relative, which is what a witness family must exploit — one wants a family so symmetric that every basis is “generic” for some member, and the rotation-invariance of the Lebesgue chain is the obvious source of such symmetry.

Acknowledgements. Our principal debt is to W. T. Gowers, for the correspondence of c. 2000 refuting the first author’s conjecture and posing Galvin’s question, and for the construction reproduced as Theorem 4.2; he declined co-authorship at the time, but the reader will see how much of §§2 and 4 is his. We thank Fred Galvin for the question answered by Theorem 2.1. Portions of these results were developed in the course of an extended dialogue with Anthropic’s Claude language model; responsibility for correctness rests with the authors.

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